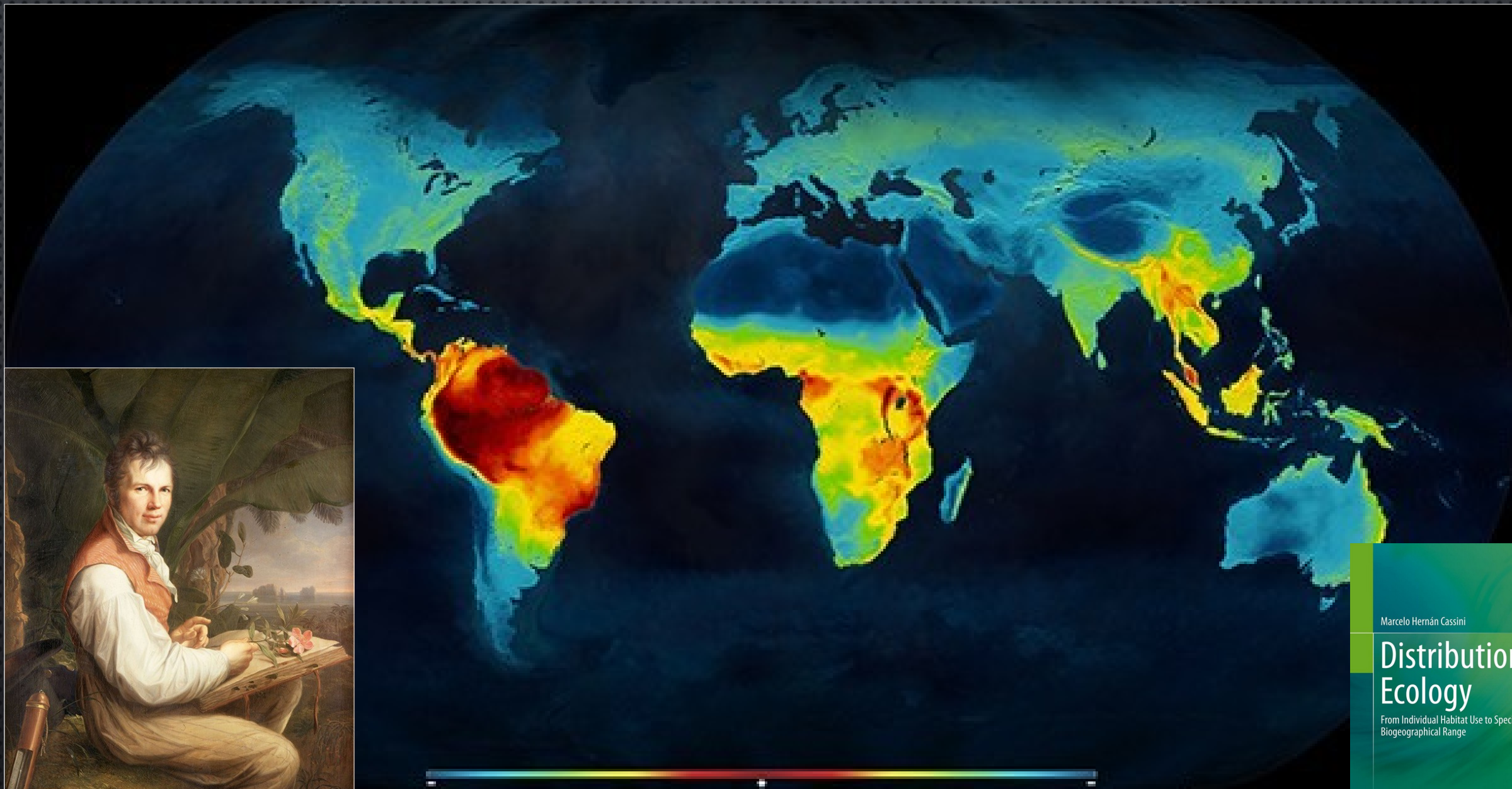


生态位模型的原理与应用（一）

THE PRINCIPLES AND APPLICATIONS OF ECOLOGICAL NICHE MODEL

中科院动物所 乔慧捷

2026-9



Marcelo Hernán Cassini

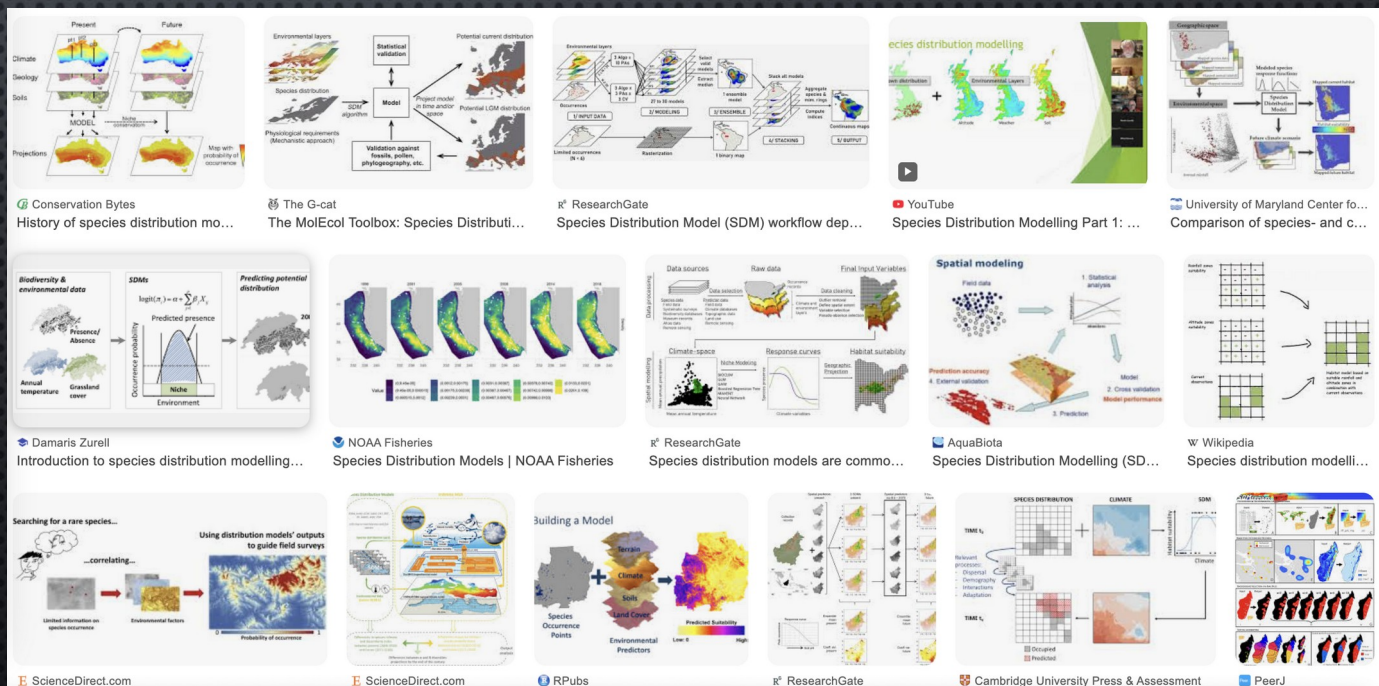
Distribution Ecology

From Individual Habitat Use to Species Biogeographical Range

什么是生态位模型？它能做些什么？

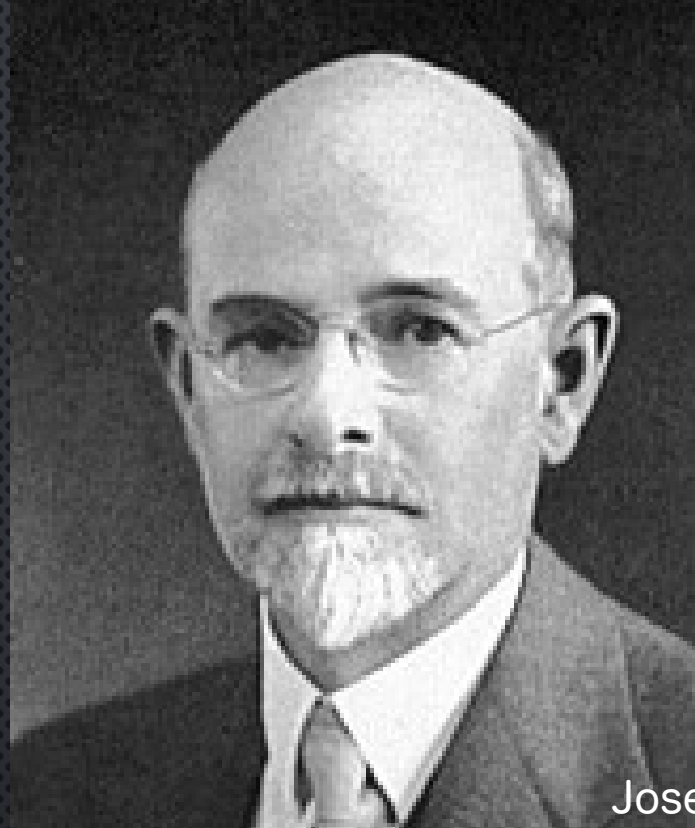
生态学和保护生物学中使用的工具，用于根据环境条件以及其它影响物种出现位置的因素来预测物种的分布

- 气候变化与物种分布
- 外来物种入侵风险分析
- 物种环境适应性
- 保护区和濒危物种保护
- 传染病传播



物种分布模型是异名

生态位概念的来源



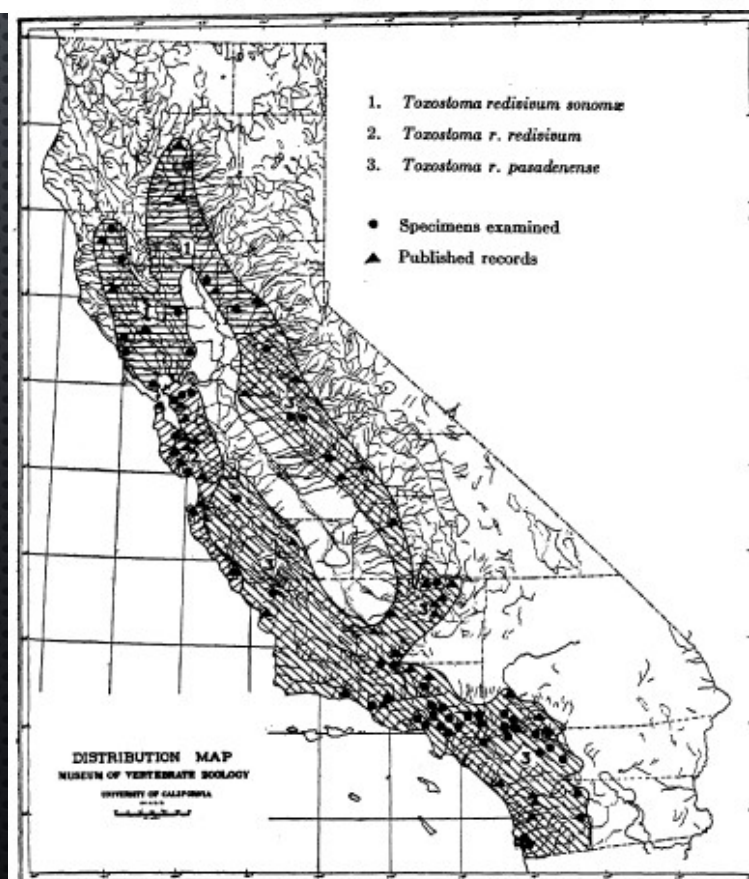
Joseph Grinnell (1877-1939)

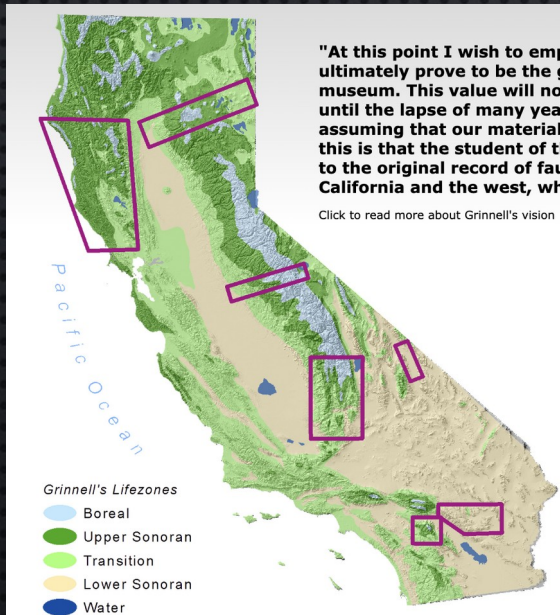
In this paper I have enumerated various factors thought to be concerned with the control of the distribution of vertebrate animals. A number of birds and mammals have been cited to show how we may use our more or less detailed knowledge of their ranges so as to demonstrate the operation of one or several out of the many possible factors as limiters to distribution. The method employed is one of examination, comparison and elimination, applied to all parts of the margin of animals' ranges. The range of any one animal must be examined at all points of its periphery in order that all of the factors concerned may be detected. One factor may constitute the barrier in one section of the periphery of the range of a species, a totally different factor in another section.

THE NICHE-RELATIONSHIPS OF THE CALIFORNIA THRASHER.¹

BY JOSEPH GRINNELL.

THE California Thrasher (*Toxostoma redivivum*) is one of the several distinct bird types which characterize the so-called "Californian Fauna." Its range is notably restricted, even more so than that of the Wren-Tit. Only at the south does the California Thrasher occur beyond the limits of the state of California, and in that direction only as far as the San Pedro Martir Mountains and





"At this point I wish to emphasize what I believe will ultimately prove to be the greatest value of our museum. This value will not, however, be realized until the lapse of many years, possibly a century, assuming that our material is safely preserved. And this is that the student of the future will have access to the original record of faunal conditions in California and the west, wherever we now work."

Joseph Grinnell, 1910

"The Uses and Methods of a Research Museum"
Popular Science Monthly

<https://mvz.berkeley.edu/Grinnell/>



Charles Sutherland Elton
(1900–1991)



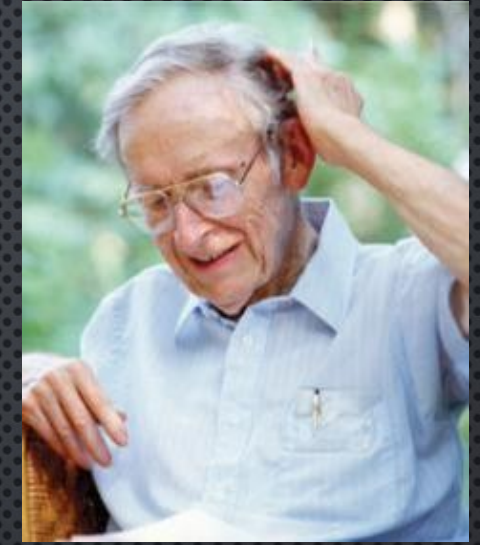
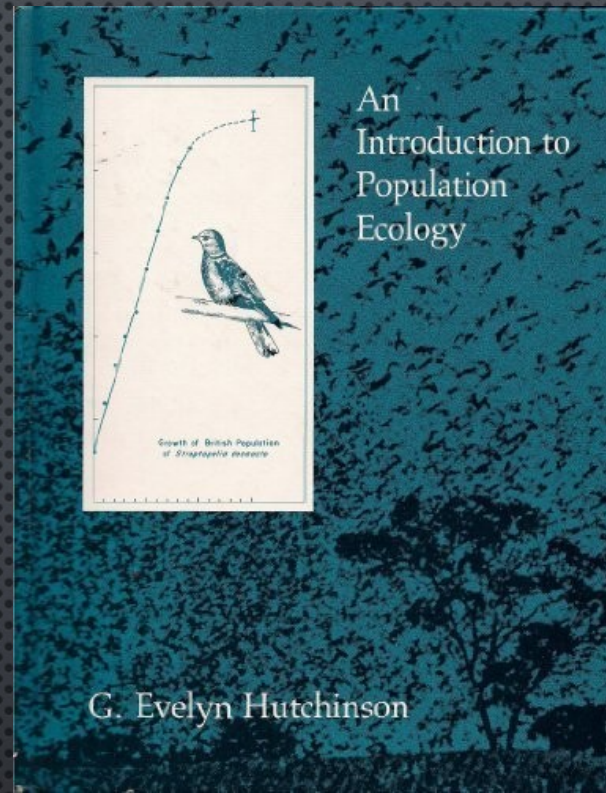
Charles Elton setting mouse traps in Bagley Wood, near Oxford, in 1926. Courtesy of Charles Elton.



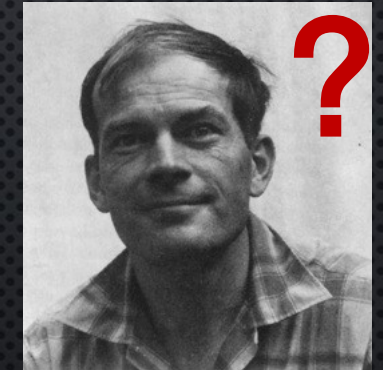
Niche as role in communities (Eltonian niche)
Animal Ecology (1927)



George Evelyn Hutchinson
(1903–1991)
Father of modern ecology



H.T. Odum and E.P. Odum



Robert MacArthur

Multidimensional ecological niches, biotic versus abiotic variables (Hutchinson Niche)
Hutchinson G E. Concluding remarks. Cold Spring, 1957, 22:415-427
An Introduction to Population Ecology (1978)

生态位概念的具像化

环境 - 分布 - 生态位空间

生态位概念的具像化

环境 - 分布 - 生态位空间

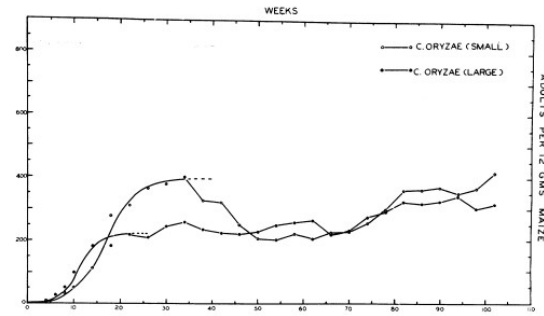


FIG. 1. The trend in numbers of adults of the small and the large strains of *C. oryzae* in separate populations in maize of 13% moisture content at 29.1° C. The smooth curves are calculated logistic curves. (See Table III.)

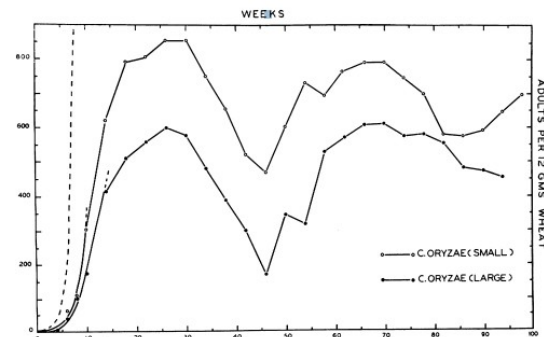
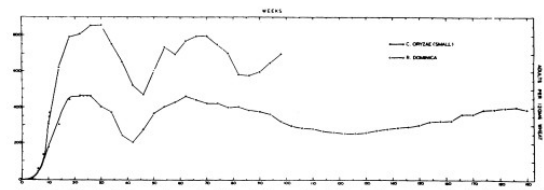


FIG. 2. The trend in numbers of adults of the small and the large strains of *C. oryzae* in separate populations in wheat of 14% moisture content at 29.1° C. The smooth curves are calculated logistic curves. (See Table III.) The broken line on the left of the diagram shows the estimated geometric increase of the small strain when $r = 0.772$.

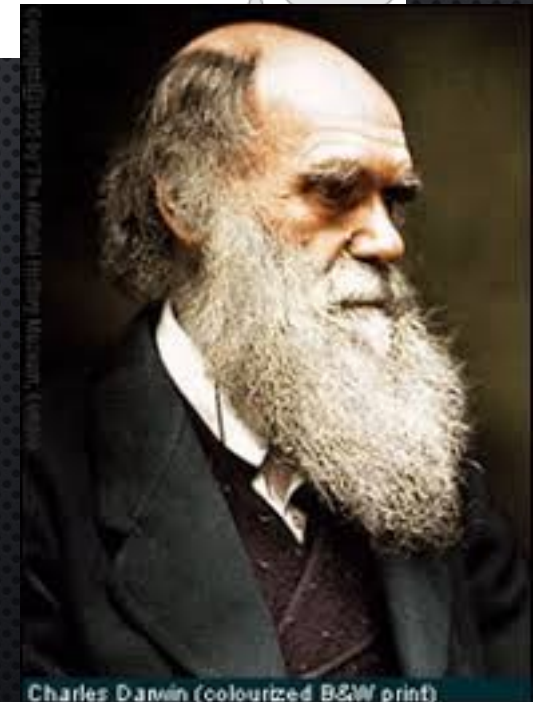
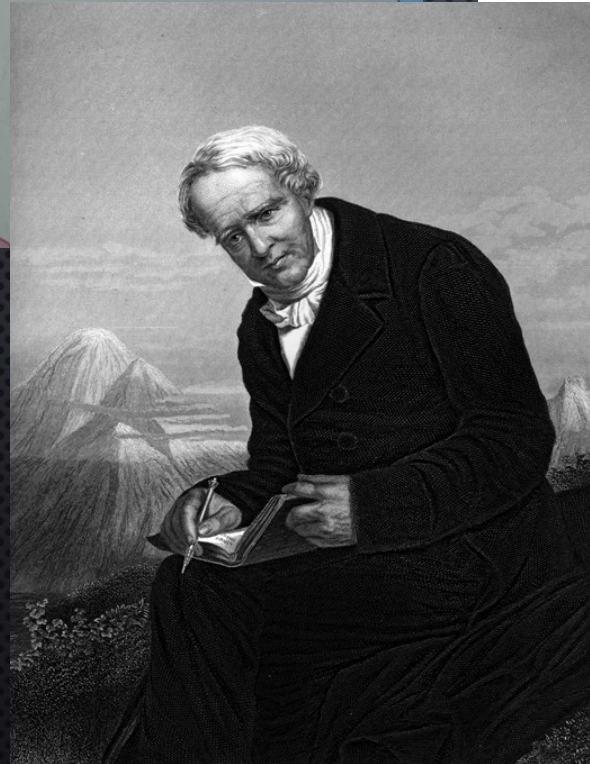
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Birch, L. C. (1953). Experimental background to the study of the distribution and abundance of insects: II. the relation between innate capacity for increase in numbers and the abundance of three grain beetles in experimental populations. *Ecology* 34(4): 712-726.

生态位概念的具像化

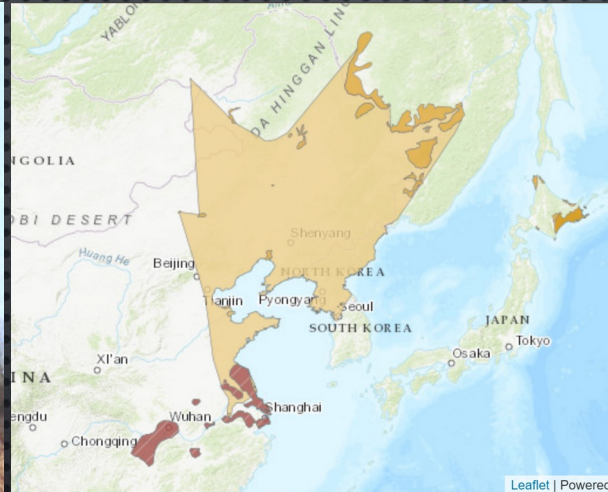
环境 - 分布 - 生态位空间



Charles Darwin (colourized B&W print)

物种的地理分布

Red-crowned Crane (丹顶鹤, *Grus japonensis*)



IUCN RedList

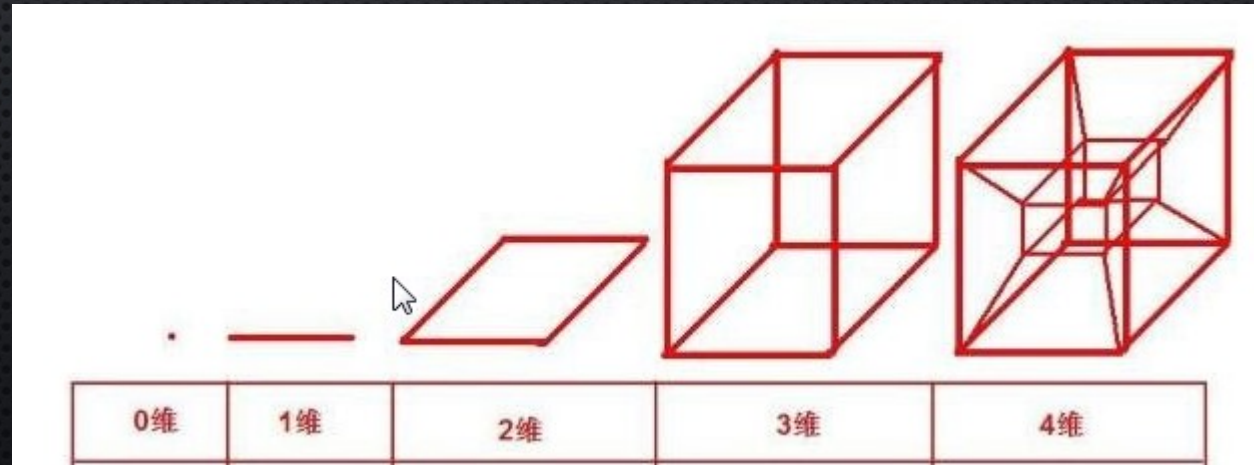
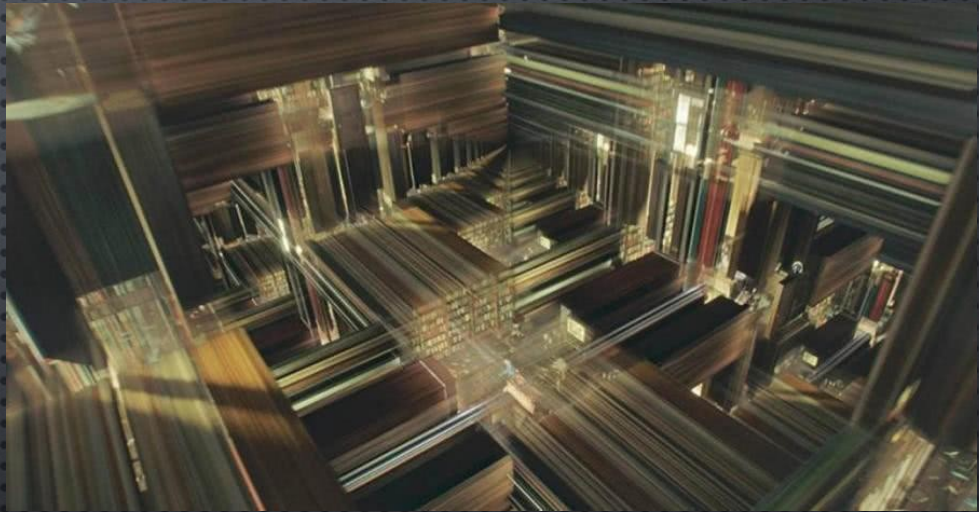
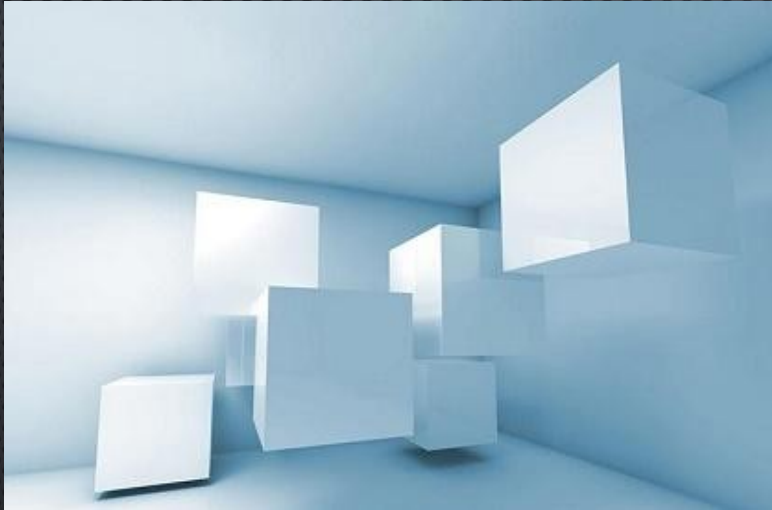


GBIF

生态位概念的具像化

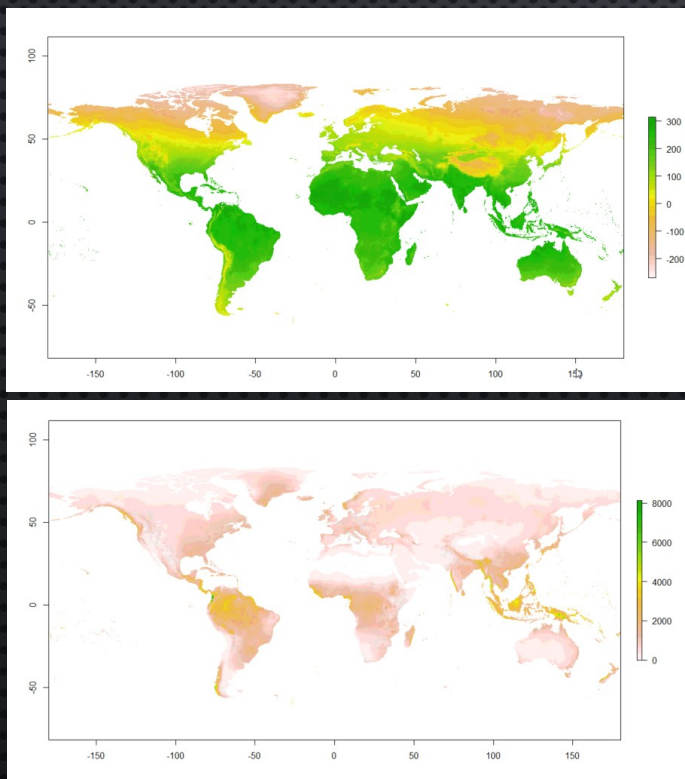
环境 - 分布 - 生态位空间

生态位空间 - 空间



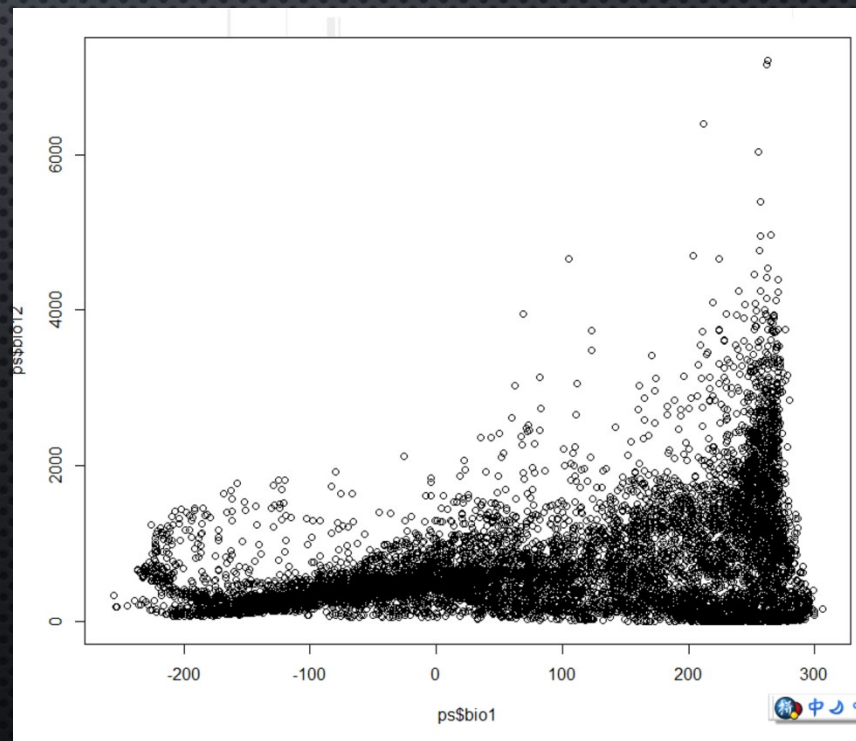
地理空间

- 以地球表面作为坐标系统
- 二维 / 三维空间
- 由纬度、经度和海拔标记
- 小尺度空间自相关
- 大尺度空间异质性

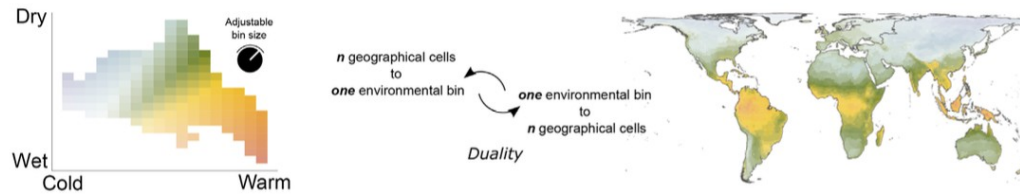


生态位空间（环境空间）

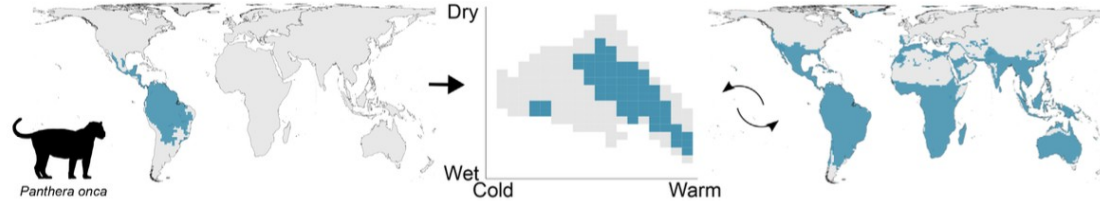
- 以环境因素为坐标系统
- 多维空间
- 非正交轴（变量之间的相关性）



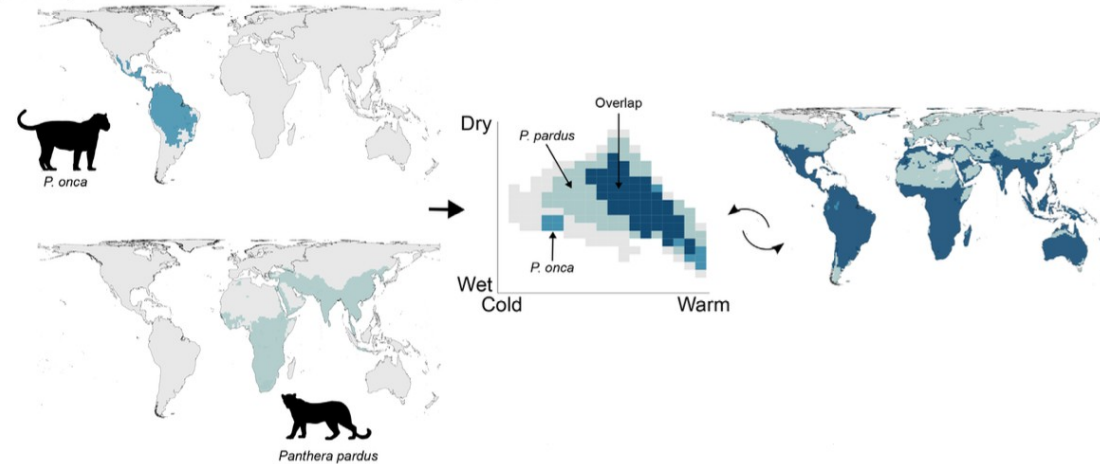
(a) Duality between environmental and geographical space



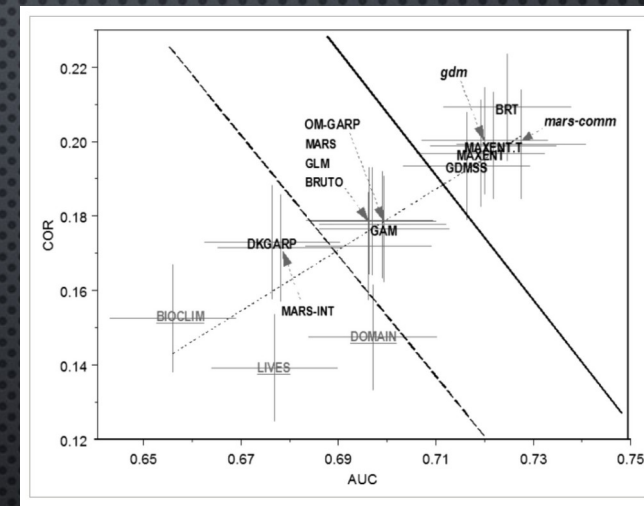
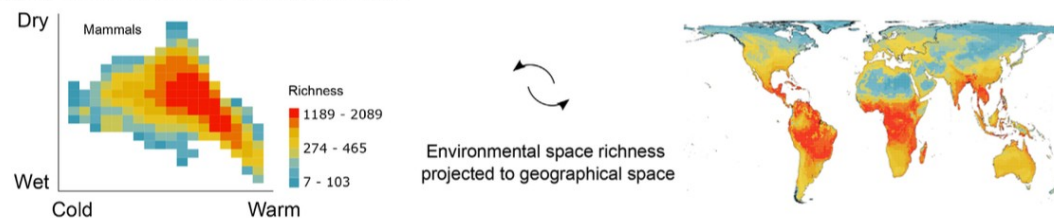
(b) Individual species attributes in environmental space



(c) Multi-species niche approaches in environmental space



(d) Mapping species richness in environmental space



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Novel methods improve prediction of species' distributions from occurrence data

Jane Elith*, Catherine H. Graham*, Robert P. Anderson, Miroslav Dudík, Simon Ferrier, Antoine Guisan, Robert J. Hijmans, Falk Huettmann, John R. Leathwick, Anthony Lehmann, Jin Li, Lucia G. Lohmann, Bette A. Loiselle, Glenn Manion, Craig Moritz, Miguel Nakamura, Yoshinori Nakazawa, Jacob McC. M. Overton, A. Townsend Peterson, Steven J. Phillips, Karen Richardson, Ricardo Scachetti-Pereira, Robert E. Schapire, Jorge Soberón, Stephen Williams, Mary S. Wisz, Niklaus E. Zimmermann ... See fewer authors

First published: 29 March 2006 | <https://doi.org/10.1111/j.2006.0906.7590.04596.x> | Citations: 6,498

J. Elith (j.elith@unimelb.edu.au), School of Botany, Univ. of Melbourne, Parkville, Victoria, 3010 Australia. — C. H. Graham, Dept of Ecology and Evolution, 650 Life Sciences Building, Stony Brook Univ., NY 11794, USA. — R. P. Anderson, City College of the City Univ. of New York, NY, USA. — M. Dudík, Princeton Univ., Princeton, NJ, USA. — S. Ferrier, Dept of Environment and Conservation Armidale, NSW, Australia. — A. Guisan, Univ. of Lausanne, Switzerland. — R. J. Hijmans, The Univ. of California, Berkeley, CA, USA. — F. Huettmann, Univ. of Alaska Fairbanks, AK, USA. — J. R. Leathwick, NIWA, Hamilton, NZ. — A. Lehmann, Suisse Centre for Faunal Cartography (CSF), Neuchâtel, Switzerland. — J. Li, CSIRO Atherton, Queensland, Australia. — L. Lohmann, Univ. de São Paulo, Brazil. — B. A. Loiselle, Univ. of Missouri, St. Louis, USA. — G. Manion, Dept of Environment and Conservation, NSW, Australia. — C. Moritz, The Univ. of California, Berkeley, USA. — M. Nakamura, Centro de Invest. en Matemáticas (CIMAT), México. — Y. Nakazawa, The Univ. of Kansas, Lawrence, KS, USA. — J. McC. Overton, Landcare Research, Hamilton, NZ. — A. T. Peterson, The Univ. of Kansas, Lawrence, KS, USA. — S. J. Phillips, AT&T Labs-Research, Florham Park, NJ, USA. — K. S. Richardson, McGill Univ., QC, Canada. — R. Scachetti-Pereira, Centro de Referência em Informação Ambiental, Brazil. — R. E. Schapire, Princeton Univ., Princeton, NJ, USA. — J. Soberón, The Univ. of Kansas, Lawrence, KS, USA. — S. E. Williams, James Cook Univ., Queensland, Australia. — M. S. Wisz, National Environmental Research Inst., Denmark. — N. E. Zimmermann, Swiss Federal Research Inst. WSL.

ECOLOGY LETTERS

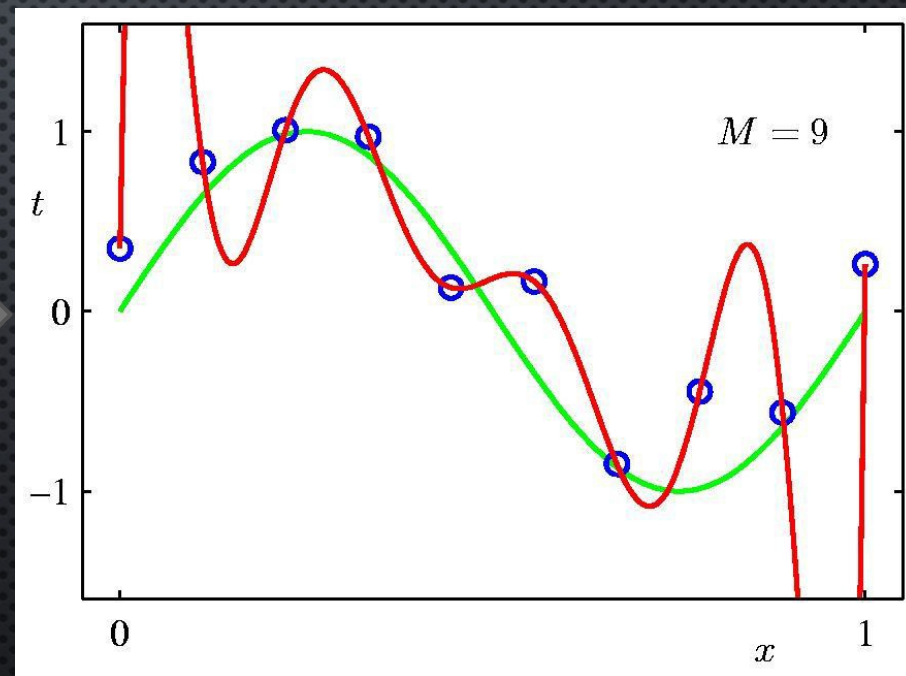
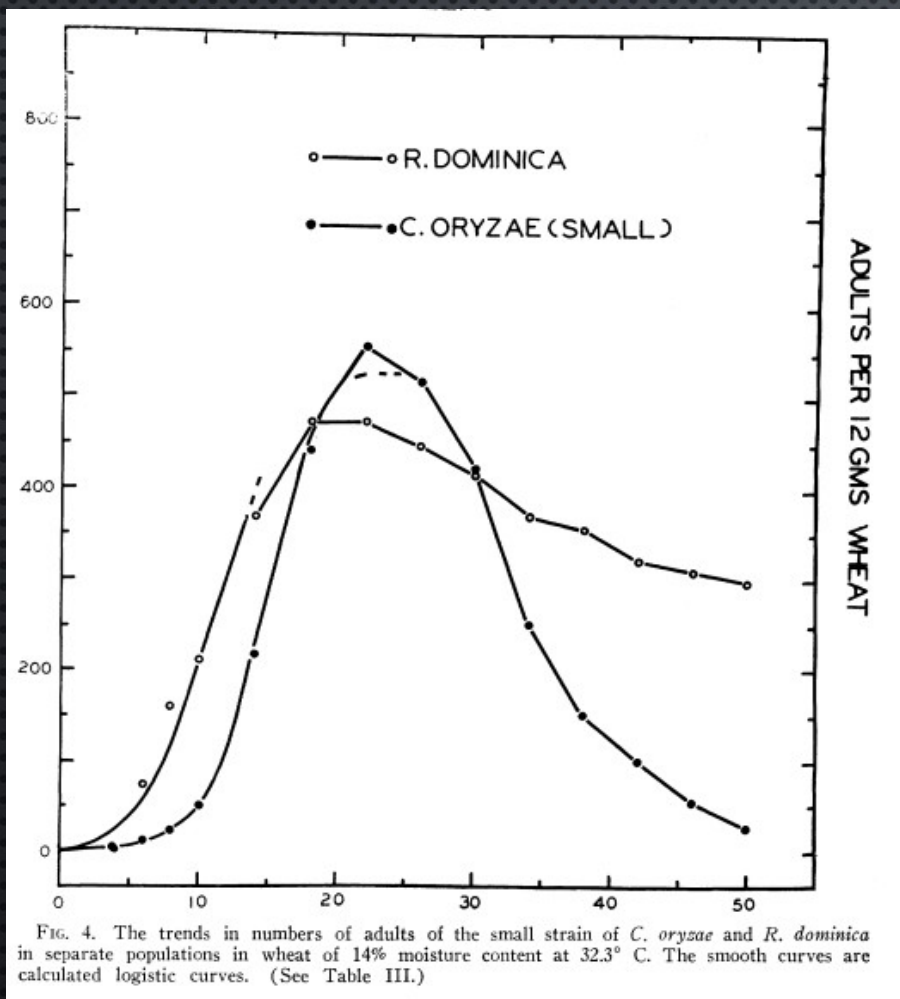
PERSPECTIVE

Biodiversity Patterns Redefined in Environmental Space

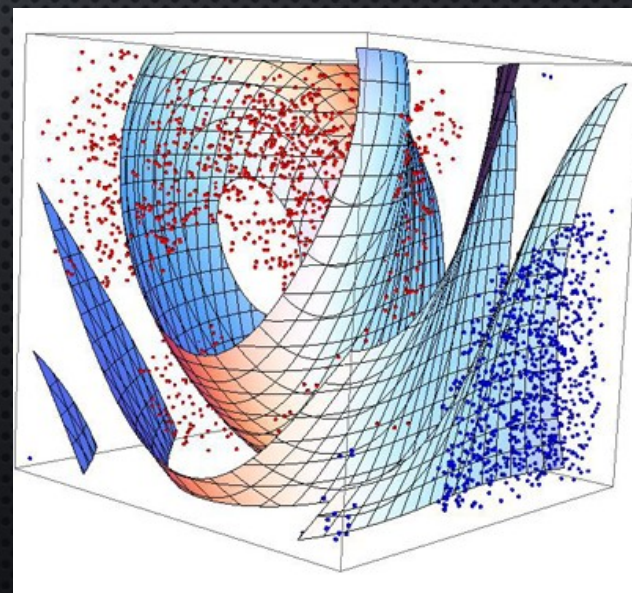
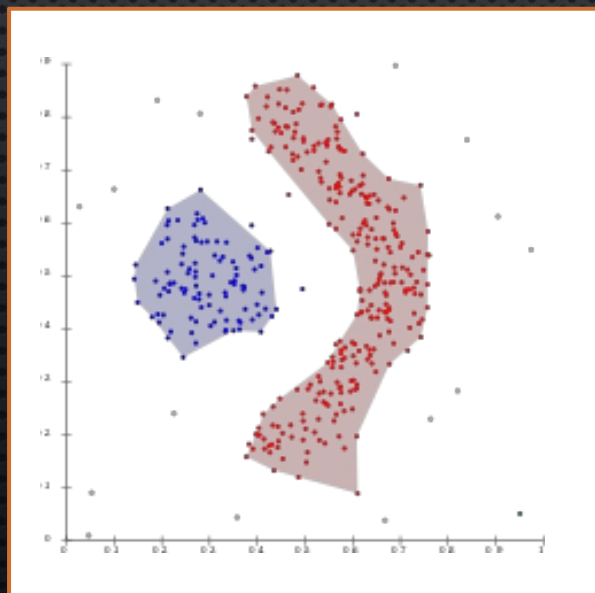
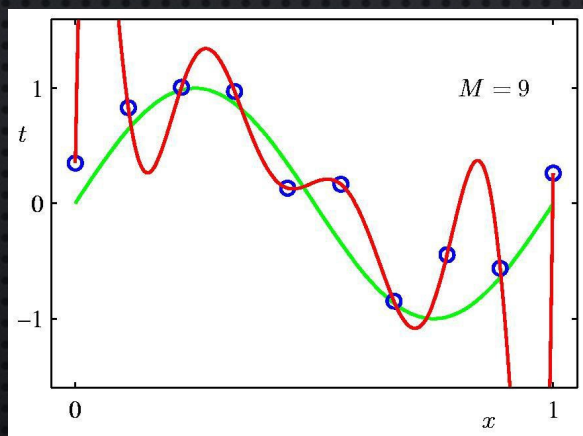
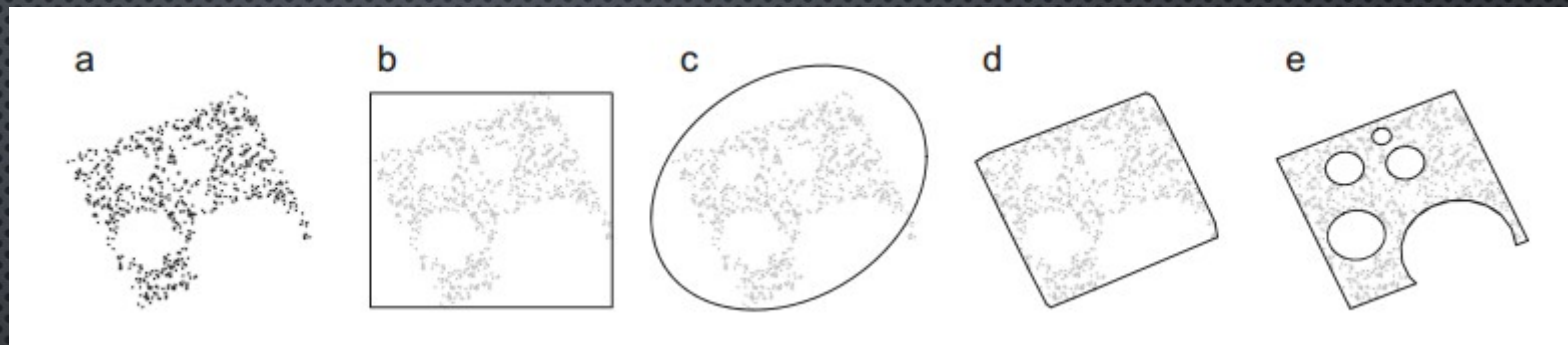
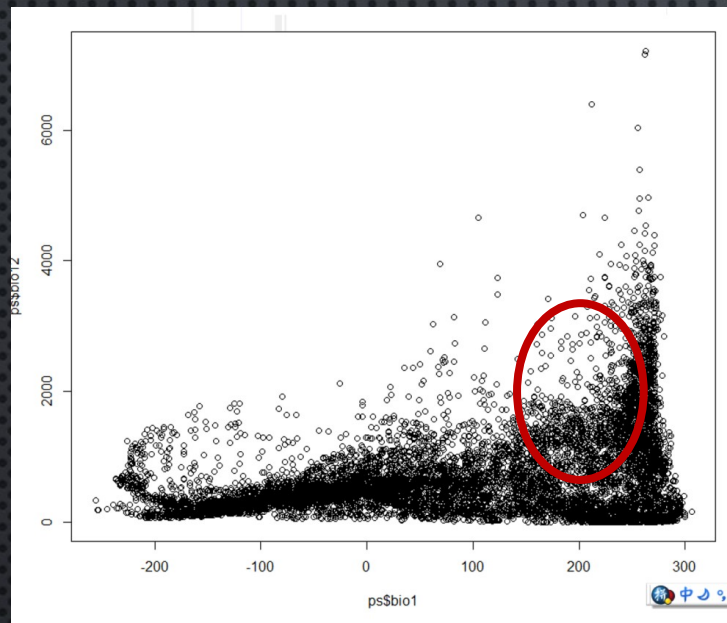
Catherine H. Graham✉, Matheus Lima Araujo, Elisa Barreto, Christian S. Dambros, José Alexandre Felizola Diniz-Filho, Niklaus E. Zimmermann, Thiago F. Rangel, Marco Túlio Pacheco Coelho

First published: 19 February 2025 | <https://doi.org/10.1111/ele.70008>

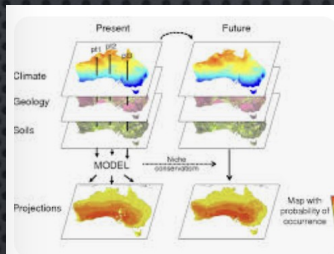
如何描述生态位?



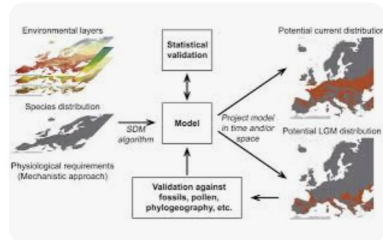
生态位在环境空间中的表现形式



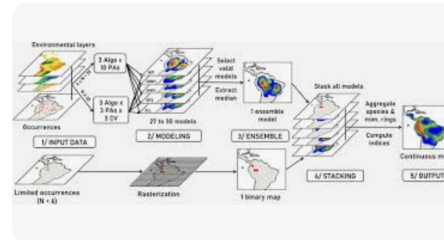
生态位建模与物种分布预测



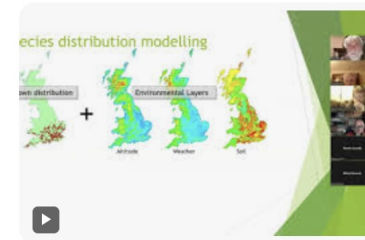
Conservation Bytes
History of species distribution mo...



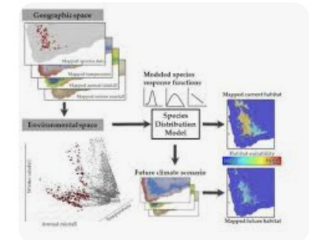
The G-cat
The MoEcol Toolbox: Species Distributi...



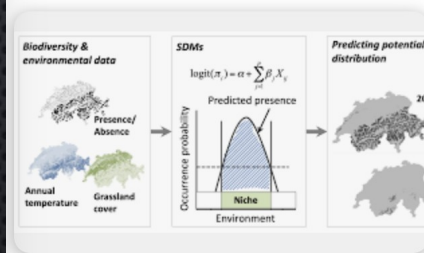
R ResearchGate
Species Distribution Model (SDM) workflow dep...



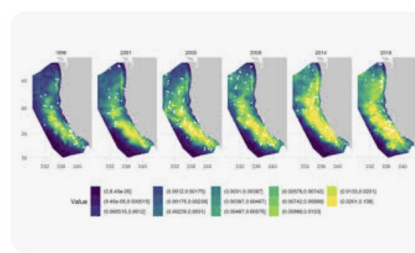
YouTube
Species Distribution Modelling Part 1: ...



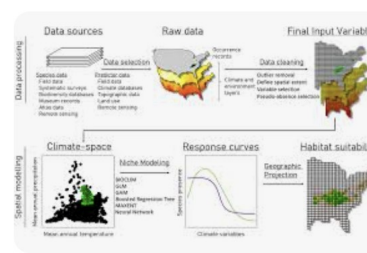
University of Maryland Center fo...
Comparison of species- and c...



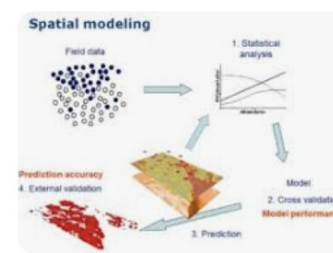
Damaris Zurell
Introduction to species distribution modelling...



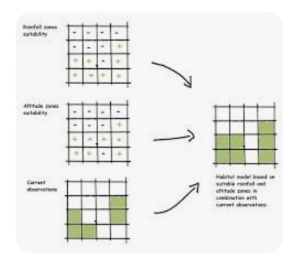
NOAA Fisheries
Species Distribution Models | NOAA Fisheries



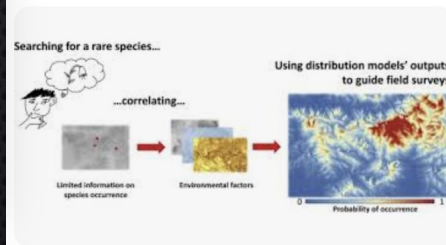
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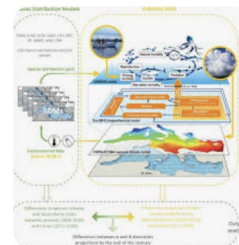
AquaBiota
Species Distribution Modelling (SD...



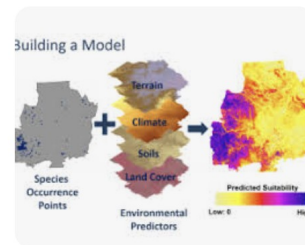
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Species distribution modelli...



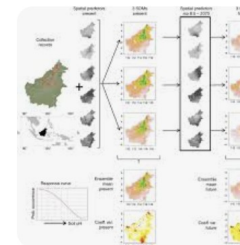
ScienceDirect.com



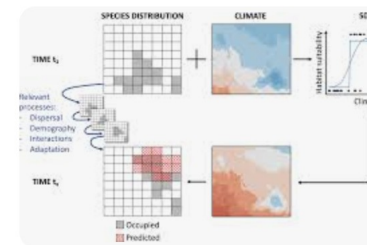
ScienceDirect.com



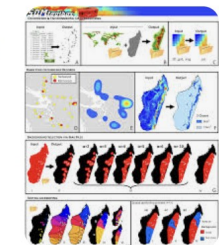
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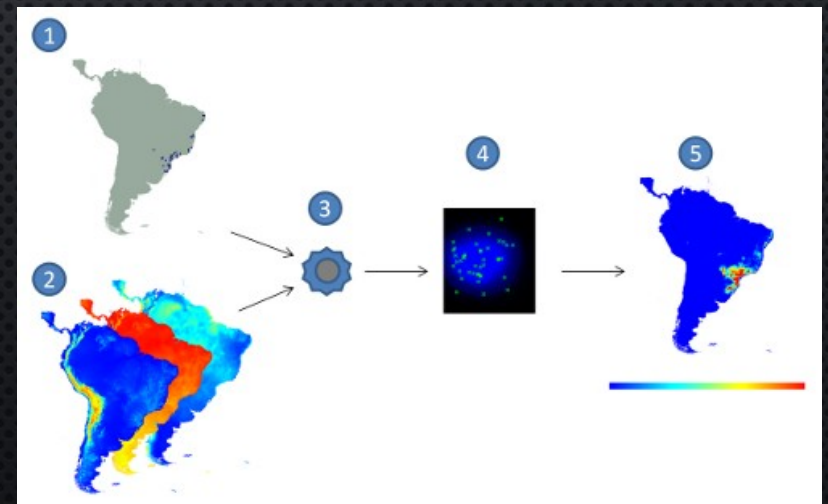
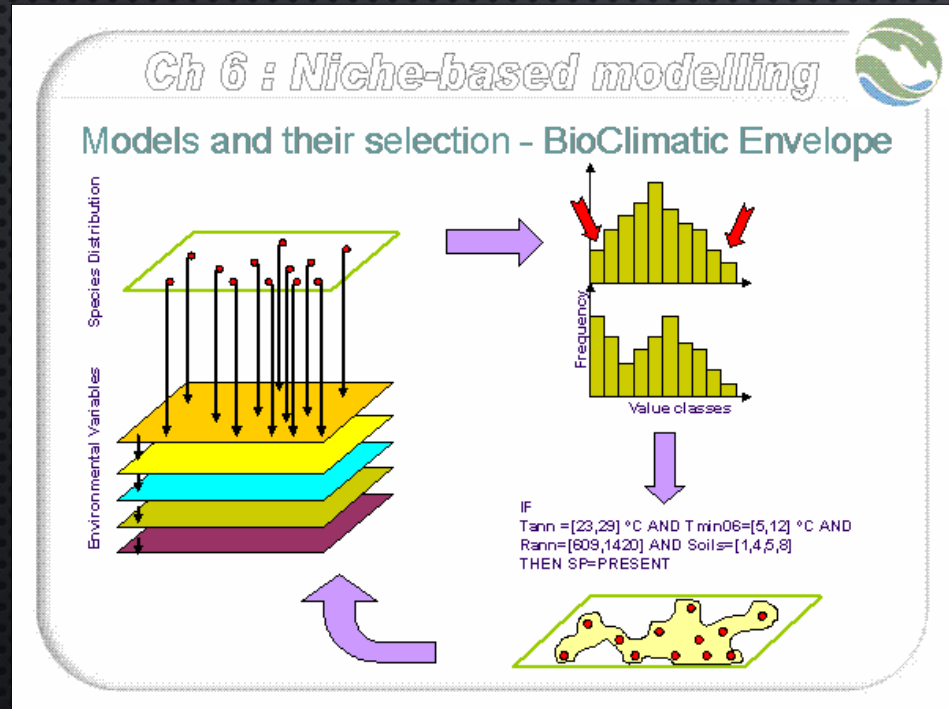
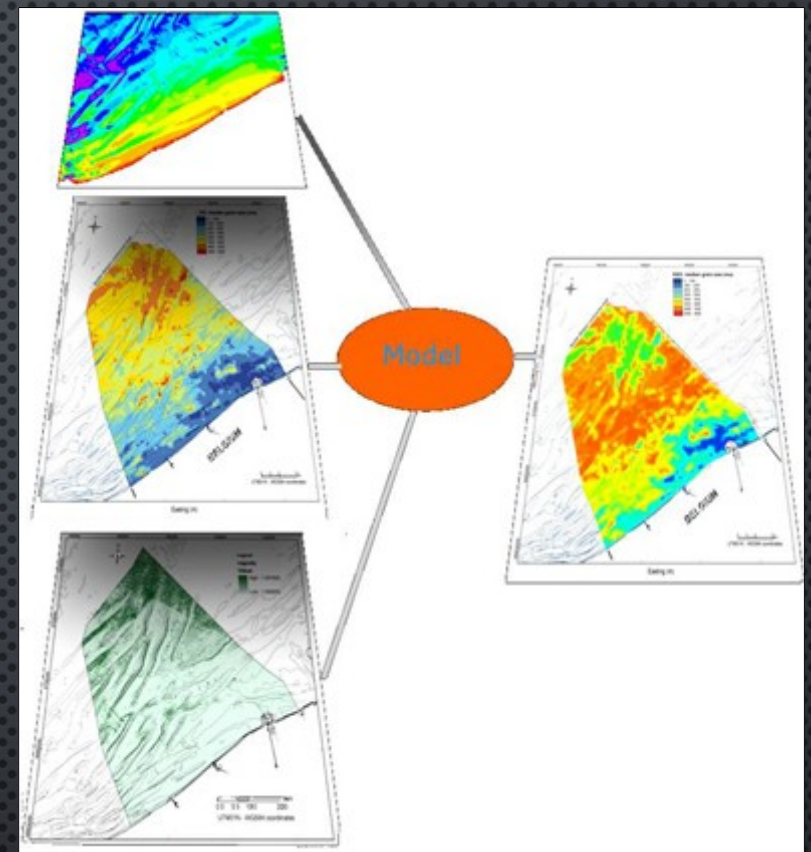
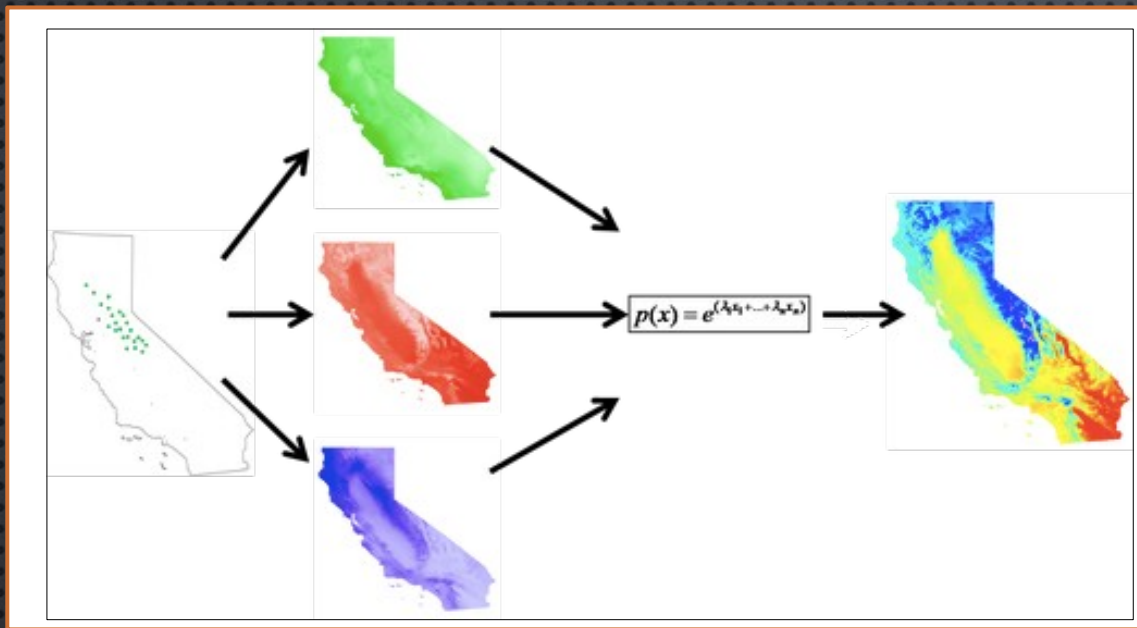
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Cambridge University Press & Assessment



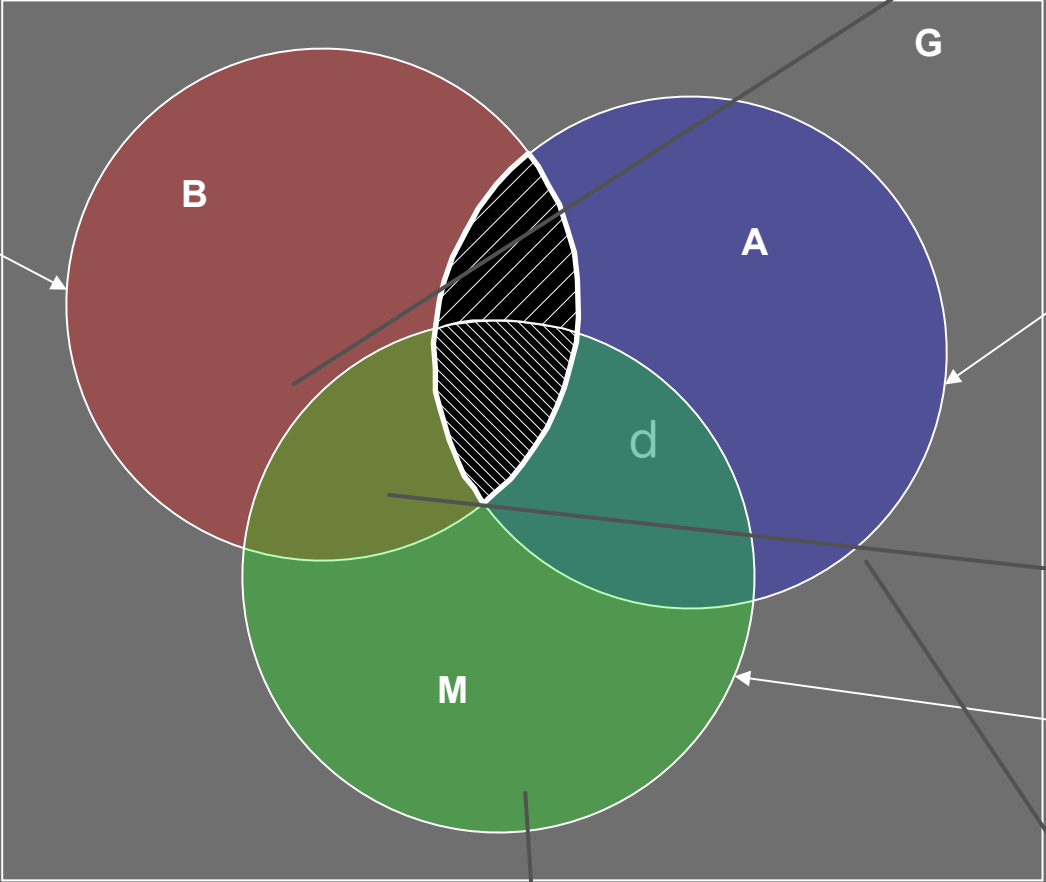
PeerJ



生态位概念定义与映射

The Area of Distribution

Favorable biotic environment
(**B**iotic)



Climate change and species distribution
Protection area and endangered species protection

Physiological requirements
(**A**biotic)

Infectious disease transmission

Accessible to dispersal
(**M**ovements)

Analysis of Invasion Risk of Alien Species

Species environmental adaptability

生态位模型建模流程

生态位建模方法

- 使用原始发生数据
- 使用栅格环境数据集
- 在模型中结合这两类数据流
- 使用该模型制作地图
- 地图被解释为物种“存在概率”



生态位建模方法

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发生数据



IUCN
RedList



GBIF

发生数据

GLOBAL BIODIVERSITY INFORMATION FACILITY (GBIF)

RED-CROWNED CRANE (丹顶鹤, *GRUS JAPONENSIS*)

[HTTPS://WWW.GBIF.ORG/](https://www.gbif.org/)

GBIF | Global Biodiversity Information Facility

Free and open access to biodiversity data

OCCURRENCES SPECIES DATASETS PUBLISHERS RESOURCES

Grus japonensis

What is GBIF? About GBIF China

Grus japonensis

EVERYTHING OCCURRENCES SPECIES DATASETS PUBLISHERS RESOURCES

Grus japonensis (Statius Muller, 1776) Species

Classification : Animalia > Chordata > Aves > Gruiformes > Gruidae > Grus

Accepted Species 5,271 occurrences



TABLE	GALLERY	MAP	TAXONOMY	METRICS	DOWNLOAD
⋮	Scientific name	Country or area	Coord		
	Grus japonensis (Statius Muller, 1776)	Russian Federation	43.9N		
	Grus japonensis (Statius Muller, 1776)	Japan	43.2N		
	Grus japonensis (Statius Muller, 1776)	Japan	43.0N		
	Grus japonensis (Statius Muller, 1776)	Japan	43.3N		
	Grus japonensis (Statius Muller, 1776)	Japan	43.2N		

DOWNLOAD OPTIONS			
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↓ DARWIN CORE ARCHIVE	✓	✓	✓ (links)
↓ SPECIES LIST	✗	✓	✗

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5,271 occurrences included in download

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RED-CROWNED CRANE (丹顶鹤, *GRUS JAPONENSIS*)

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id	countryCode	locality	stateProvince	occurrenceStatus	individualCount	publishingOrgKey	decimalLatitude	decimalLongitude
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	JP	弟子屈町--川湯温泉 (Teshikaga--Kawayu Hot Spring)	Hokkaido	PRESENT	NA	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.63738	144.44035
	CN	CH-Zhalong Nature Reserve	Heilongjiang	PRESENT	8	e2e717bf-551a-4917-bdc9-4fa0f342c530	47.20873	124.23858
	CN	CH-Zhalong Nature Reserve	Heilongjiang	PRESENT	15	e2e717bf-551a-4917-bdc9-4fa0f342c530	47.20873	124.23858
	JP	釧路市--釧路湿原 (Kushiro City--Kushiro Wetlands) (general)	Hokkaido	PRESENT	23	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.09169	144.40602
	CN	向海国家级自然保护区 (Xianghai NNR)	Jilin	PRESENT	1	e2e717bf-551a-4917-bdc9-4fa0f342c530	45.02285	122.34816
	KR	강원도 철원 & 비무장지대(DMZ--Cheorwon Basin General ...	Kang-won-do	PRESENT	NA	e2e717bf-551a-4917-bdc9-4fa0f342c530	38.22739	127.27661
	JP	根室市--春国岱原生野鳥公園 (Nemuro--Shunkunitai Wild Bi...	Hokkaido	PRESENT	2	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.27320	145.46065
	JP	別海町--野付半島--ナラワラ (Betsukai--Notsuke Peninsula-...	Hokkaido	PRESENT	2	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.60897	145.27477
	JP	Hokkaido--Tokotan bridge	Hokkaido	PRESENT	4	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.48347	145.24892
	JP	釧路市--釧路市阿寒国際ツルセンター (Kushiro City--Kushir...	Hokkaido	PRESENT	2	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.14339	144.14919
	CN	Yellow River Nature Reserve	Shandong	PRESENT	3	e2e717bf-551a-4917-bdc9-4fa0f342c530	37.98859	118.07831
	JP	釧路市--釧路市丹頂鶴自然公園 (Kushiro--Red-crowned Cra...	Hokkaido	PRESENT	NA	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.05794	144.19656
	JP	釧路市--釧路市丹頂鶴自然公園 (Kushiro--Red-crowned Cra...	Hokkaido	PRESENT	NA	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.05794	144.19656
	CN	Dongying -- Yellow River Mouth	Shandong	PRESENT	1	e2e717bf-551a-4917-bdc9-4fa0f342c530	37.79387	118.95996
	JP	釧路市--釧路市阿寒国際ツルセンター (Kushiro City--Kushir...	Hokkaido	PRESENT	104	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.14339	144.14919
	JP	鶴居村--音羽橋 (Tsurui--Otowa Bridge)	Hokkaido	PRESENT	48	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.18774	144.33421
	JP	Akkeshi Region (highway)	Hokkaido	PRESENT	NA	e2e717bf-551a-4917-bdc9-4fa0f342c530	43.10549	144.88426

发生数据

GLOBAL BIODIVERSITY INFORMATION FACILITY (GBIF)

RED-CROWNED CRANE (丹顶鹤, *GRUS JAPONENSIS*)

[HTTPS://WWW.GBIF.ORG/](https://www.gbif.org/)

```
library(data.table)
setwd("C:/Users/Huijie Qiao/Dropbox/Meetings/讲座/20241114 - 亚洲鸟会")
df<-fread("0006777-241107131044228.csv")
df<-df[, c("species", "decimalLongitude", "decimalLatitude")]
df<-df[!is.na(decimalLongitude) & !is.na(decimalLatitude)]
colnames(df)<-c("species", "x", "y")
fwrite(df, "occ.csv")
```

	species	x	y
1	Grus japonensis	144.89680	43.04551
2	Grus japonensis	144.44035	43.63738
3	Grus japonensis	124.23858	47.20873
4	Grus japonensis	124.23858	47.20873
5	Grus japonensis	144.40602	43.09169
6	Grus japonensis	122.34816	45.02285
7	Grus japonensis	127.27661	38.22739
8	Grus japonensis	145.46065	43.27320
9	Grus japonensis	145.27477	43.60897
10	Grus japonensis	145.24892	43.48347
11	Grus japonensis	144.14919	43.14339
12	Grus japonensis	118.07831	37.98859
13	Grus japonensis	144.19656	43.05794
14	Grus japonensis	144.19656	43.05794
15	Grus japonensis	118.95996	37.79387
16	Grus japonensis	144.14919	43.14339
17	Grus japonensis	144.33421	43.18774
18	Grus japonensis	144.88426	43.10549
19	Grus japonensis	141.46483	43.00544

生态位建模方法

- 使用原始发生数据
- 使用栅格环境数据集
- 在模型中结合这两类数据流
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- 地图被解释为物种“存在概率”

栅格环境数据集

BIOCLIMATIC VARIABLES

HTTPS://WWW.WORLDCCLIM.ORG/DATA/BIOCLIM.HTML

 WorldClim

Bioclimatic variables

Bioclimatic variables are derived from the monthly temperature and rainfall values in order to generate more biologically meaningful variables. These are often used in species distribution modeling and related ecological modeling techniques. The bioclimatic variables represent annual trends (e.g., mean annual temperature, annual precipitation) seasonality (e.g., annual range in temperature and precipitation) and extreme or limiting environmental factors (e.g., temperature of the coldest and warmest month, and precipitation of the wet and dry quarters). A quarter is a period of three months (1/4 of the year).

They are coded as follows:

BIO1 = Annual Mean Temperature

BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))

BIO3 = Isothermality (BIO2/BIO7) (×100)

Historical climate data

Historical monthly weather data

Future climate data

Below you can download the standard (19) WorldClim [Bioclimatic variables](#) for WorldClim version 2. They are the average for the years 1970-2000. Each download is a "zip" file containing 19 GeoTiff (.tif) files, one for each month of the [variables](#).

variable	10 minutes	5 minutes	2.5 minutes	30 seconds
Bioclimatic variables	bio 10m	bio 5m	bio 2.5m	bio 30s

名称	修改日期	类型	大小
 bio1.asc	2015/2/11 13:59	ASC 文件	10,205 KB
 bio2.asc	2015/2/11 13:59	ASC 文件	10,029 KB
 bio3.asc	2015/2/11 13:59	ASC 文件	9,679 KB
 bio4.asc	2015/2/11 13:59	ASC 文件	10,984 KB
 bio5.asc	2015/2/11 13:59	ASC 文件	10,210 KB
 bio6.asc	2015/2/11 13:59	ASC 文件	10,397 KB
 bio7.asc	2015/2/11 13:59	ASC 文件	10,242 KB
 bio8.asc	2015/2/11 13:59	ASC 文件	10,129 KB
 bio9.asc	2015/2/11 14:00	ASC 文件	10,390 KB
 bio10.asc	2015/2/11 13:59	ASC 文件	10,149 KB
 bio11.asc	2015/2/11 13:59	ASC 文件	10,405 KB
 bio12.asc	2015/2/11 13:59	ASC 文件	10,330 KB
 bio13.asc	2015/2/11 13:59	ASC 文件	9,879 KB
 bio14.asc	2015/2/11 13:59	ASC 文件	9,433 KB
 bio15.asc	2015/2/11 13:59	ASC 文件	9,755 KB

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ENM/SDM ≡ MAXENT?

- [HTTPS://BIODIVERSITYINFORMATICS.AMNH.ORG/OPEN_SOURCE/MAXENT/](https://BIODIVERSITYINFORMATICS.AMNH.ORG/OPEN_SOURCE/MAXENT/)



Maximum Entropy Species Distribution Modeling, Version 3.3.3k

Samples
File: **Browse**

Environmental layers
Directory/File: **Browse**

☒ Linear features
☒ Quadratic features
☒ Product features
☒ Threshold features
☒ Hinge features
☒ Auto features

☐ Create response curves
☒ Make pictures of predictions
☐ Do jackknife to measure variable importance

Output format: **Logistic**
Output file type: **asc**

Output directory: **Browse**
Projection layers directory/file: **Browse**

Run **Settings** **Help**

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MAXENT

• [HTTPS://BIODIVERSITYINFORMATICS.AMNH.ORG/OPEN_SOURCE/MAXENT/](https://BIODIVERSITYINFORMATICS.AMNH.ORG/OPEN_SOURCE/MAXENT/)

Maximum Entropy Species Distribution Modeling, Version 3.3.3k

Samples

Filesgs\讲座\20241114 - 亚洲鸟会\occ.cBrowse

☒ Grus_japonensis

Environmental layers

Directory/Filetings\讲座\20241114 - 亚洲鸟会\bioclimBrowse

☒ bio1

Continuous

☒ bio10

Continuous

☒ bio11

Continuous

☒ bio12

Continuous

☒ bio13

Continuous

☒ bio14

Continuous

☒ bio15

Continuous

☒ bio16

Continuous

☒ bio17

Continuous

☒ bio18

Continuous

☒ bio19

Continuous

Select all

Deselect all

☒ Linear features

☒ Quadratic features

☒ Product features

☒ Threshold features

☒ Hinge features

☒ Auto features

Create response curves☒

Make pictures of predictions☒

Do jackknife to measure variable importance☒

Output formatLogistic

Output file typeasc

Output directorybox\Meetings\讲座\20241114 - 亚洲鸟会\maxent.fullBrowse

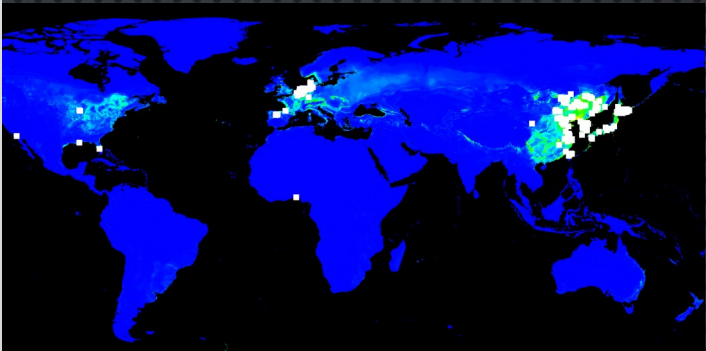
Projection layers directory/fileBrowse

Run

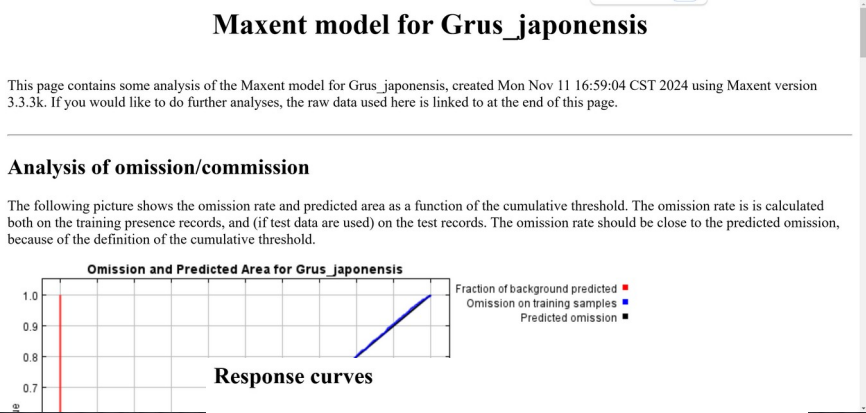
Settings

Help

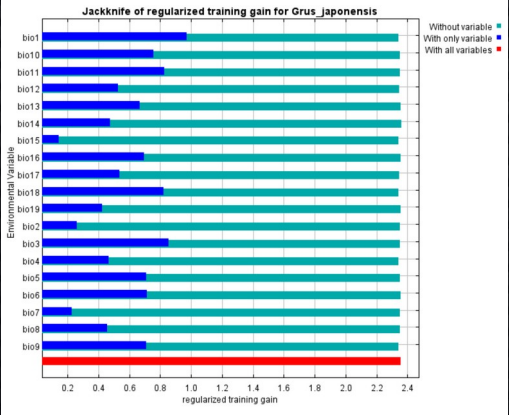
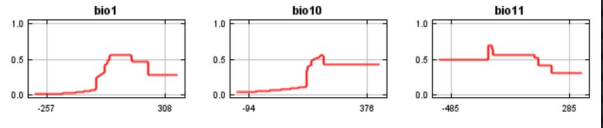
plots	2024/11/11 16:59
Grus_japonensis.asc	2024/11/11 16:59
Grus_japonensis.html	2024/11/11 16:59
Grus_japonensis.lambdas	2024/11/11 16:59
Grus_japonensis_omission.csv	2024/11/11 16:59
Grus_japonensis_sampleAverages.csv	2024/11/11 16:59
Grus_japonensis_samplePredictions.c...	2024/11/11 16:59
maxent.log	2024/11/11 16:59
maxentResults.csv	2024/11/11 16:59



Variable	Percent contribution	Permutation importance
bio18	30.9	14.2
bio3	23.4	2.7
bio10	15.2	7.7
bio1	6	19.8
bio17	3.4	3
bio13	3.4	1.7
bio5	2.9	0.9



These curves show how each environmental variable affects the Maxent prediction. The curves show the response curve to see a larger version. Note that the curves can be hard to interpret if you have strong collinearity between variables, whereas the model may take advantage of sets of variables changing together.



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很简单，对不？